SE Lab 7

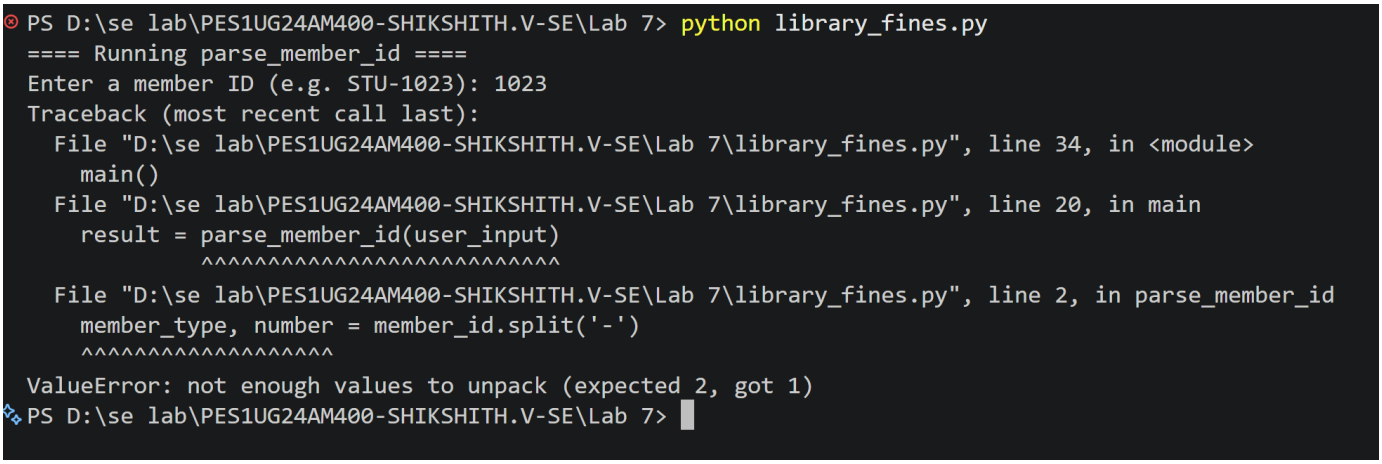
Fuzzing for Input Validation Bugs

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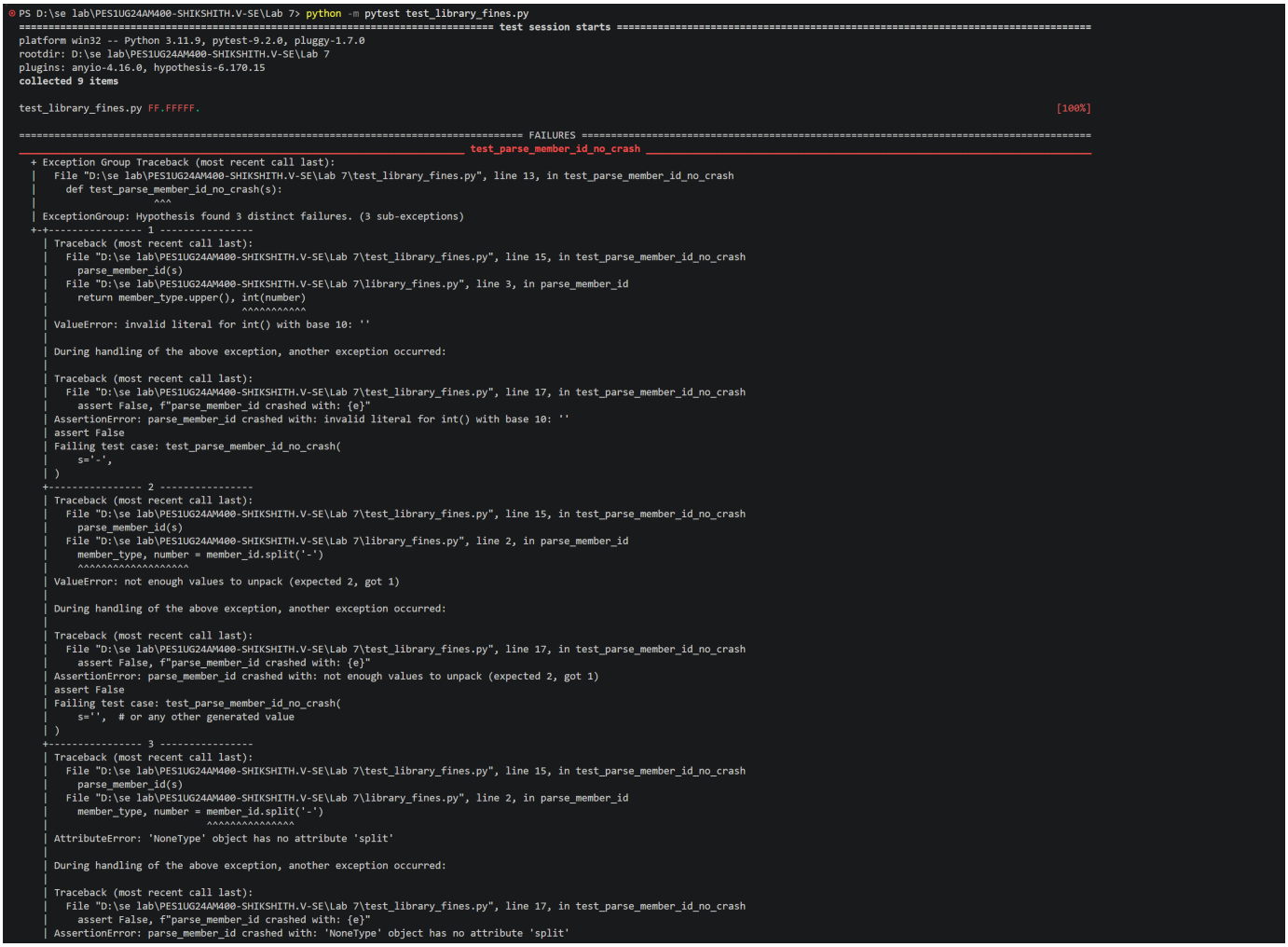
Step 2: Original buggy application (python library_fines.py)

Entering a member ID without the STU- prefix (1023) crashes parse_member_id with a ValueError.



Step 4: Hypothesis fuzz tests failing on the buggy code

pytest collects 9 tests; Hypothesis finds failing inputs such as '', '-' and None.



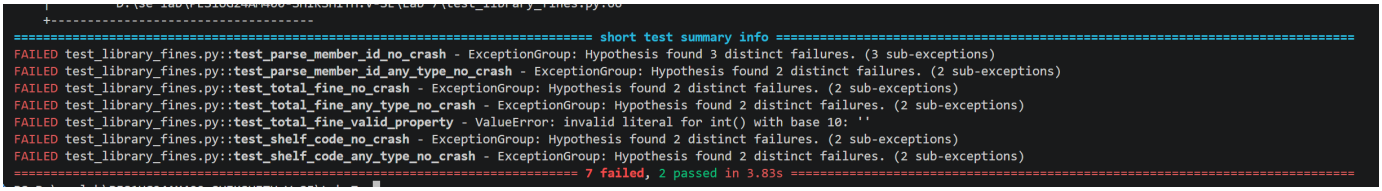
```
PS D:\se lab\PES1UG24AM400-SHIKSHITH.V-SE\Lab 7> python -m pytest test_library_fines.py
============================= test session starts ==============================
platform win32 -- Python 3.11.9, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\se lab\PES1UG24AM400-SHIKSHITH.V-SE\Lab 7
plugins: anyio-4.16.0, hypothesis-6.170.15
collected 9 items

test_library_fines.py FF.FFFFF.                                          [100%]

================================== FAILURES ===================================
_____________________ test_parse_member_id_no_crash ______________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\se lab\PES1UG24AM400-SHIKSHITH.V-SE\Lab 7\test_library_fines.py", line 13, in test_parse_member_id_no_crash
  |     def test_parse_member_id_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\se lab\PES1UG24AM400-SHIKSHITH.V-SE\Lab 7\test_library_fines.py", line 15, in test_parse_member_id_no_crash
    |     parse_member_id(s)
    |   File "D:\se lab\PES1UG24AM400-SHIKSHITH.V-SE\Lab 7\library_fines.py", line 3, in parse_member_id
    |     return member_type.upper(), int(number)
    |                                 ^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\se lab\PES1UG24AM400-SHIKSHITH.V-SE\Lab 7\test_library_fines.py", line 17, in test_parse_member_id_no_crash
    |     assert False, f"parse_member_id crashed with: {e}"
    | AssertionError: parse_member_id crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_parse_member_id_no_crash(
    |     s='-',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\se lab\PES1UG24AM400-SHIKSHITH.V-SE\Lab 7\test_library_fines.py", line 15, in test_parse_member_id_no_crash
    |     parse_member_id(s)
    |   File "D:\se lab\PES1UG24AM400-SHIKSHITH.V-SE\Lab 7\library_fines.py", line 2, in parse_member_id
    |     member_type, number = member_id.split('-')
    |     ^^^^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\se lab\PES1UG24AM400-SHIKSHITH.V-SE\Lab 7\test_library_fines.py", line 17, in test_parse_member_id_no_crash
    |     assert False, f"parse_member_id crashed with: {e}"
    | AssertionError: parse_member_id crashed with: not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_parse_member_id_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "D:\se lab\PES1UG24AM400-SHIKSHITH.V-SE\Lab 7\test_library_fines.py", line 15, in test_parse_member_id_no_crash
    |     parse_member_id(s)
    |   File "D:\se lab\PES1UG24AM400-SHIKSHITH.V-SE\Lab 7\library_fines.py", line 2, in parse_member_id
    |     member_type, number = member_id.split('-')
    |                           ^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\se lab\PES1UG24AM400-SHIKSHITH.V-SE\Lab 7\test_library_fines.py", line 17, in test_parse_member_id_no_crash
    |     assert False, f"parse_member_id crashed with: {e}"
    | AssertionError: parse_member_id crashed with: 'NoneType' object has no attribute 'split'
```

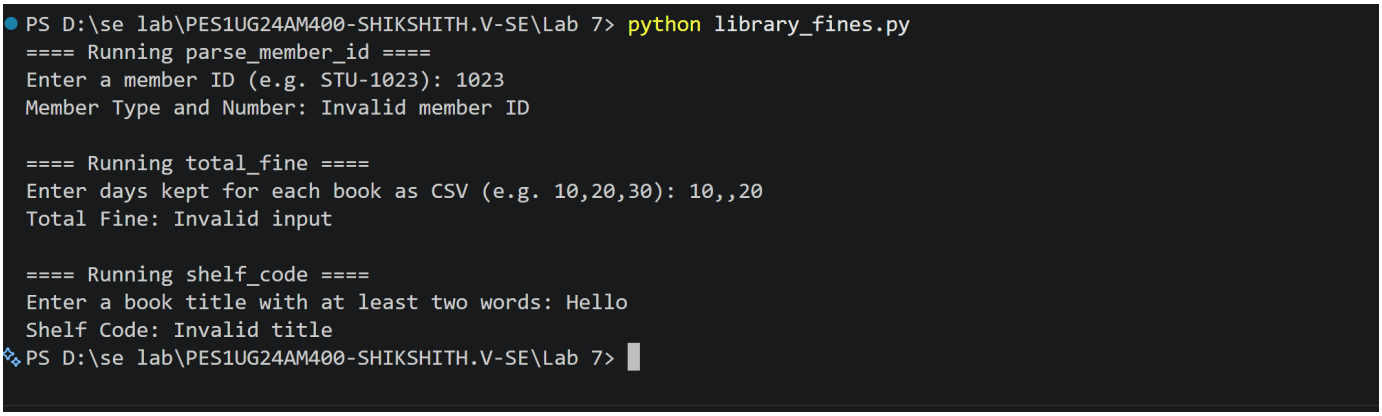
Step 4: Failure summary

7 failed, 2 passed. The two passing tests only use valid inputs.



Step 5: Fixed application (python library_fines.py)

The same bad inputs (1023, 10,,20, Hello) are now rejected with clear messages and no crash.



Step 6: Fuzz tests re-run on the fixed code

All 9 tests pass.

